THE UNIVERSITY OF TEXAS AT DALLAS

CS4485.0w1 – Computer Science Project

Fall 2026

Project Weekly Report

Project Title	Air Pollution Analyzer and Predictor
Faculty Advisor	Muhammad Ikram
Group Number (E.g., Ikram_T1_30)	Ikram_T7_36
Student Names	Chris Abellana, Aldo Carmona, Alexander Constante, Andy Do, Faiz Khan, Yael Rico
Group/Team Lead	Alexander Constante
Reporting Week (E.g., 9/14/26 to 9/18/26)	10/12/2026 to 10/16/2026
Meeting Attendees (students, faculty advisor, TA, etc.)	Students

Note. Submit completed report before next meeting. Add more bullets if needed.

1. Progress made in the reporting week (list all tasks who worked on it, and time spent on each task):
    a. Training ML model using XGBoost with base aqi dataset - Yael (40 mins)
    b. Data research and compilation - Andy (45 mins)
    c. Created simple front end - Aldo (45 mins)

2. Challenges faced in the reporting week:
    a. Finding data set that monitors traffic density
    b. Other commitments (Classes, Jobs, etc.)
    c. Lack of group focus on singular tasks (Spread of interest)

3. Plans for next week:
    a. Clean up base data set and discuss other feature implementations
    b. Pick other ML algorithms and compare accuracy metrics over them
    c. Getting data on how air quality affects health, especially for those who are more sensitive to air quality changes
    d. Begin implementing the front end.